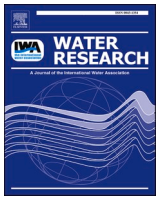
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Methylmercury photodegradation in paddy water: An overlooked process mitigating methylmercury risks

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ABSTRACT

Photodegradation is critical to reduce the potent neurotoxic methylmercury (MeHg) in water and its subsequent accumulation along food chains. However, this process has been largely ignored in rice paddies, which are hotspots of MeHg production and receive about a quarter of the world's developed freshwater resources. Here, we reported that significant MeHg photodegradation, primarily mediated by hydroxyl radicals, occurs in the overlying water during rice growth. By incorporating field-measured light interception into a rice paddy biogeochemistry model, as well as photodegradation rates obtained from 42 paddy soils stretching ~3500 km across China, we estimated that photodegradation reduced MeHg concentrations in paddy water and rice by 82 % and 11 %, respectively. Without photodegradation, paddy water could be a significant MeHg source for downstream ecosystems, with an annual export of 178 – 856 kg MeHg to downstream waters in China, the largest rice producer. These findings suggest that photodegradation in paddy water is critical for preventing greater quantities of MeHg entering human food webs.

1. Introduction

Methylmercury (MeHg) is one of the most toxic forms of mercury (Hg) (Wang et al., 2019) and almost all humans are exposed to this potent neurotoxin via consuming aquatic products and rice (Boening, 2000). Once absorbed, MeHg causes irreversible neurological damage and harms cognitive function (Giang and Selin, 2016). Dietary exposure to MeHg is reported to decrease intelligence quotient (IQ) by 12,000,000 points and is responsible for 29,000 deaths per year globally, resulting in an economic loss of about $US 117 billion (Zhang et al., 2021). The primary source of MeHg bioaccumulation in human food webs is water, while photodegradation is the dominant pathway of MeHg removal from natural waters (Hammerschmidt and Fitzgerald, 2006; Seller et al., 1996). Dissolved organic matter (DOM) and reactive oxygen species (ROS), particularly hydroxyl radicals and singlet oxygen, have been reported to dominate MeHg photodegradation in surface waters (Li et al., 2024; Sun et al., 2015; Zhang and Hsu-Kim, 2010). Through photodegradation, MeHg levels in water columns can be effectively reduced, which subsequently lowers MeHg accumulation in organisms. For example, it has been reported that photodegradation removed ~83 % of MeHg in Toolik Lake (Lehnherr and St. Louis, 2009) and up to 85 % in polar glaciers (Girard et al., 2016).

To date, MeHg photodegradation has been ignored in rice-paddy systems, which are wetlands receiving 24 – 30 % of the world's developed freshwater resources (Global Rice Science Partnership, 2013). This is partially due to the commonly held view that rice leaves intercept solar radiation (Alpers et al., 2014), substantially weakening MeHg photodegradation in the surface paddy water. As a result, the extent to which MeHg photodegradation occurs in rice-paddy systems remains unknown. Additionally, as hotspots of Hg methylation,

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rice-paddy systems nourish rice as a dominant source of MeHg exposure for populations consuming rice as a staple food, e.g., accounting for up to 96 % of total MeHg exposure for Asians (Gong et al., 2018). Rice-paddy systems could also be a significant source of MeHg for downstream lakes and rivers via runoff (Brigham et al., 2009), with subsequent impacts on MeHg accumulation in aquatic food webs and human exposure. It is, therefore, of great importance to quantify how MeHg photodegradation affects MeHg concentrations in paddy water and rice. However, the properties of paddy water that impact MeHg photodegradation (Black et al., 2012), such as pH and DOM content, are distinct from those of natural waters due to farming activities. These differences make it challenging to extrapolate MeHg photodegradation in lakes and oceans to the paddy water. To solve this problem, it is necessary to quantify MeHg photodegradation under light interception by rice plants and in paddy water with various properties, as well as develop a model that could link this process with MeHg concentrations in paddy water and rice.

The main objective of this study is to elucidate the importance of photodegradation in MeHg removal from rice-paddy systems. To achieve that, MeHg photodegradation in paddy water and light interception by rice plants were firstly quantified throughout the entire rice growth period by conducting pot experiments, while rice plant-induced light interception was monitored in a field survey. Then, 42 types of paddy soils collected from 12 major rice-producing provinces in China were used in laboratory simulation experiments to broadly quantify photodegradation rates and to investigate the driving force of MeHg photodegradation in paddy water. Based on these data, a refined rice paddy biogeochemical model (RPBM) (Kwon et al., 2018) was introduced to quantitatively assess the impacts of photodegradation on MeHg accumulation in rice and its export to downstream environments. The results obtained from this study are essential for enhancing our understanding of MeHg photodegradation in rice-paddy systems and provide new insights into mitigating MeHg risks for both rice consumers and downstream aquatic ecosystems.

2. Materials and methods

2.1. Experimental design

Pot experiment. A pot experiment was carried out to explore MeHg photodegradation in paddy surface water during the period of rice growth. Specifically, rice plants were cultivated in 6 pots, and each pot contained 4 kg of paddy soil collected from a paddy field in Nanjing, Jiangsu Province (referred to as JSNJ). Concentrations of soil total Hg (THg) and MeHg were 185.8 ± 11.1 and 0.88 ± 0.07 μg kg^{-1}, respectively. Pots were arranged in several rows and planted with three to five plants (Wufeng 2168) in each pot to match the planting density observed in the field and to ensure equal spacing between plants. All pots were placed in an outdoor ground under natural light conditions, with an ambient temperature ranging from 11 to 38 ℃. On Day 30 (transplanting day), 60 (tillering and jointing stage), 90 (heading and flowering stage), and 130 (filling and maturity stage) of rice growth, three quartz tubes, each containing 30 mL of paddy water spiked with MeHg to an environmentally relevant level of 10 ng/L (Zhang et al., 2018) were placed under each of the rice plants. Paddy water was obtained by suspending JSNJ soil in deionized water at a ratio of 1:1 (w/w) in an incubator shaker (25 ℃, 120 r min^{-1}) for 1 h, and then filtered through a 0.22-μm membrane. Three quartz tubes without any shading were also set up as controls. MeHg adsorption on the wall of quartz tubes has been quantified and found to be insignificant. All the quartz tubes were separated from the overlying water and soils and sealed to avoid water loss from evaporation and interference from the ambient environment (Fig. S1a, Supporting Information, SI). Photosynthetically active radiation (PAR) was monitored (Photosynthetically active radiometer, GLZ-C, China) during 48-hour experimental periods. In each experimental period, after 0, 12, 24, 36, and 48 h of incubation with natural illumination, 3 mL of sample from each quartz tube was collected, acidified with hydrochloric acid to 0.4 % (Han et al., 2017), and subjected to the measurement of MeHg.

Field survey. To further explore the effects of light interception by rice plants during the whole growth period and correct the photodegradation rates in model evaluation, PAR was monitored at different stages of rice growth in two paddy fields in Hainan Province. Hainan province was chosen to measure the PAR in real paddy fields mainly because Hainan has rice plants at different growth stages within a certain time and measuring PAR at different growth stages within a couple of days can minimize the impact of weather. The PAR on days 30, 60, and 90 were measured in the same field and then in another field ~50 km away on day 120 of rice growth. The ratio of PAR measured with rice plants to that without any shading was defined as the interception rate in this study.

Preparation of paddy water for laboratory simulation experiments. The paddy water of each paddy soil was prepared by adding deionized water to the soil and equilibrating for 1 h in an incubator shaker (25 ℃, 120 r min^{-1}). Water samples were then filter-sterilized (0.22 μm) to rule out microbial demethylation. Laboratory incubation instead of in situ collection of paddy water was adopted to avoid interference from local environmental factors, such as precipitation (Dai et al., 2021). Specifically, 42 types of subsurface paddy soils (2 − 15 cm) were collected from non-Hg contaminated areas after rice harvest and then immediately transported to the laboratory in ice boxes. These soils were distributed across 12 provinces (Table S1), accounting for 83 % of the total rice planting area in China (Chinese Statistical Yearbook 2021). After being brought back to the laboratory, the soils were air-dried and then sieved through a 100-μm mesh. A proportion of soil was used to measure MeHg concentration for the model evaluation. A proportion of each aqueous sample was subjected to the measurement of paddy water properties, including pH, dissolved organic carbon (DOC), dissolved anion concentrations (i.e., Cl^-, NO_3^-, SO_4^{2-}), and light transmittance, while the remaining volumes were stored in a refrigerator at 4 ℃ before use in the laboratory simulation experiments within 2 days to avoid significant aggregation of DOM.

Laboratory simulation experiments. The effects of reactive species, pH, and light quality on MeHg photodegradation were explored via laboratory simulations. Details of the experimental design are listed in Supplementary Table 2. Specifically, 1) for active species, the roles of hydrated electrons (E_{aq}^-), hydroxyl radicals (•OH), singlet oxygen ($^1O_2$), and superoxide anions ($O_2^{\bullet-}$) were examined. Quenchers of these different reactive species were separately added into the paddy waters of HBHG, HLJQTH, and JSCZ soils (Table S2). The types and concentrations of quenchers were chosen according to Han et al. (2017), under which the impacts of quenchers on MeHg determination were minor. 2) For pH, three pH levels i.e., 4, 7, and 10 (adjusted with KOH solution (50 %, w/w) or H_2SO_4 solution (pH = 2)), were examined in 3 types of paddy water (i.e., HBHG, HLJQTH, and JSCZ, which vary in pH or TOC levels, Table S2) due to the detected pH range for 42 paddy waters (4.1 − 9.5; Table S1). 3) For light quality, the roles of UVA and UVB were explored in 6 types of paddy water (i.e., GZQN, GZZY, JSCZ, JSNJ, JXJA, and SXXY, distributed in different regions of China). Before illumination, quartz tubes containing paddy water were wrapped with UVB-block film or UV-block film, referred to as UVB-block and UV-blocked treatments, respectively. Quartz tubes without any wrapping were referred to as full-spectrum radiation and PAR was measured during the whole experimental period. All paddy waters were spiked with MeHg and equilibrated in the dark for 1 h, then exposed to 500 W xenon lamps for 6 h (for soil properties, active species, and pH) or natural light (for light quality). After illumination, 3 mL of samples were collected and acidified with HCl before MeHg analysis. The generation of hydroxyl radicals was tested in pH- and light quality-treatments.

Chemical analysis. MeHg in paddy water was extracted according to the GEOTRACES method and analyzed based on EPA Method 1630. Briefly, 2 mL of the sample was digested in a 40 mL amber glass vial with

35 mL of 2 % H_2SO_4 for ~12 h. After digestion, the sample was neutralized with 2 mL of 50 % KOH and then buffered to pH = 4 − 5 with 2 M acetate buffer. 40 μL of $NaBEt_4$ was then added to the sample to generate volatile species for analysis, and MeHg was determined using an automated Hg analyzer (Brooks Rand Model III, USA). Soil MeHg was extracted using 25 % KOH—CH_3OH (w/w) and analyzed based on EPA Method 1630 (Tang et al., 2023). Quality control of MeHg analysis included triplicate sample analysis, method/reagent blanks, matrix spikes, and the analysis of certified reference material. The recoveries of certified reference material and matrix spikes ranged from 85 to 115 %, and the relative standard deviations of triplicate analyses were all < 20 %.

To provide further evidence of the role of hydroxyl radicals in MeHg photodegradation, the generations of hydroxyl radicals in paddy waters with different pH and UVA radiation were quantified. The quantification of hydroxyl radicals was based on the measurement of the product (i.e., hydroxybenzoic acid) of hydroxyl radicals reacting with benzoic acid (BA) using high-performance liquid chromatography (HPLC) (Liu et al., 2016). Hydroxybenzoic acid was separated using a mobile phase of 80 % acetonitrile and 20 % water at 1 mL min^{-1} and detected using UV absorbance at 255 nm for the p-isomer and 300 nm for the o- and m-isomers. The concentration factor used to convert total hydroxybenzoic acid from detected p-isomer formation rate was 5.87 according to a previous study (Zhou and Mopper, 1990), and the consumption of BA (C_t/C_0) was used to indicate the generation of hydroxyl radicals.

For other parameters (e.g., pH, TOC, dissolved anion, and light transmittance) of paddy water: pH was measured by a portable pH meter (HACH HQ30D, USA); the dissolved anions were measured by ion chromatography (Dionex ICS900, USA); DOC was measured using a total organic carbon analyzer (Elementar Vario TOC cube, Germany), and light transmittance was quantified by UV–vis spectrophotometer (Varian Palo Alto, USA).

2.2. Model simulation

To evaluate the impacts of MeHg photodegradation on rice MeHg accumulation and MeHg flux in a rice-paddy system, we modified the rice paddy biogeochemical model (RPBM) developed by Kwon et al. (2018). To incorporate MeHg photodegradation rates into the model, the photodegradation rates under natural light (k_{PD}, $m^2\ E^{-1}$) were calculated based on the observed photodegradation rates (k_D, h^{-1}) in 42 types of paddy water in laboratory simulation experiments. Then the photodegradation rates during the rice growth period (k_{Pt}, $m^2\ E^{-1}$) were estimated according to the PAR at different growth stages obtained from the field survey. The model was then calibrated and evaluated according to observed MeHg concentrations in paddy soils (See 2.1) and rice grains (details below). Finally, the MeHg photodegradation process was removed from the RPBM to estimate the MeHg levels in the absence of photodegradation. By comparing the differences in output MeHg levels with and without photodegradation, its impacts on MeHg concentrations in rice grains and in surface water could be evaluated.

k_{PD}, k_D, and **k_{Pt}.** k_{PD} is the photodegradation rate under natural light conditions. Given the MeHg photodegradation experiment was carried out for 6 types of paddy water in laboratory simulation experiments (light-quality treatments), the k_{PD}s of these samples could be calculated using the equation (Lehnherr and St. Louis, 2009):

$$k_{PD} = -\frac{d[MeHg]}{dt} \Big/ ([MeHg] \times PAR) \quad (1)$$

Where *[MeHg]* is the concentration of MeHg in water (ng L^{-1}), k_{PD} is the photodegradation constant of MeHg ($m^2\ E^{-1}$), and *PAR* is the photosynthetically active radiation ($E\ m^{-2}\ s^{-1}$) measured in the laboratory simulation experiments (light quality).

For the remaining 36 types of paddy water, only photodegradation rates were quantified under constant illumination, i.e., only k_Ds were available. Therefore, k_{PD} was estimated by correcting k_D for PAR (Girard et al., 2016). Briefly, the k_{PD}s obtained from 6 soils were regressed with the k_D obtained from laboratory simulation experiments under constant illumination. Given the observed k_{PD} was well correlated to the calculated k_{PD} ($r^2 = 0.95$, $p < 0.05$, Fig. S2), this relationship, together with the k_D values, was applied to calculate k_{PD} values for the remaining 36 types of paddy water. The calculation of k_D was according to Eq. (2) (Li et al., 2010; Tai et al., 2014):

$$k_D = \ln\left([MeHg]_0 / [MeHg]t\right)/t \quad (2)$$

Where $[MeHg]_0$ indicates the initial MeHg concentration and $[MeHg]_t$ is the MeHg concentration after *t*-h of illumination.

Following, the photodegradation rate during rice planting, k_{Pt} (i.e., actual photodegradation rate in rice fields with shading), could be calculated for each site. Specifically, the k_{Pt} was corrected for light interception at each growth stage. Assuming that rice plants grow linearly during each stage (Meng et al., 2011), the k_{Pt} could be calculated according to equations 3 − 7:

$$k_{Pt}/k_{PD} = 1 - 0.0078t (0 < t \leq 30) \quad (3)$$

$$k_{Pt}/k_{PD} = 1.194 - 0.0145t (30 < t \leq 60) \quad (4)$$

$$k_{Pt}/k_{PD} = 0.4602 - 0.0022t (60 < t \leq 90) \quad (5)$$

$$k_{Pt}/k_{PD} = 0.0039t - 0.0939 (90 < t \leq 120) \quad (6)$$

$$k_{Pt}/k_{PD} = 0.3765 (t > 120) \quad (7)$$

Wherein each equation represents the inhibition rates obtained from field surveys according to time (*t*).

Model modification and evaluation. Details of model description and related coefficients can be found in Kwon et al. (2018), Supplementary text 1, and Table S3. Three main modifications were made to the original model: 1) Instead of using a constant degradation rate for the whole country, we applied provincial-specific k_{PD} to the modified RPBM. 2) Instead of employing a constant photodegradation rate for the whole rice growing period, we applied the shading effect-corrected value, i.e., k_{Pt}. 3) We also used the province-specific MeHg concentrations in paddy water to explore regional differences.

Soil MeHg and rice MeHg concentrations were selected to evaluate the modified model and the model evaluation was performed at a provincial scale. For soil MeHg concentrations, we applied the detected soil MeHg concentrations for these 42 soils to better evaluate the modified model (details in 2.1). For rice MeHg concentrations, we combine the data reported by Xu et al. (2020) and our national survey (unpublished data) for each site. The data used for model evaluation are listed in Table S3. Although the MeHg concentration in paddy surface water is an important parameter, it was not used in the model evaluation because the model outputs the total MeHg (kg) content in each pool, and MeHg concentration in the surface water largely depends on the total water mass, which changes with the water depth.

To further emphasize the importance of MeHg photodegradation to downstream aquatic systems, the mass of MeHg runoff from rice-paddy systems was compared with atmospheric Hg deposition. Hg deposition was chosen considering that it is an important source of Hg in aquatic systems and is a major contributor to MeHg that accumulates in aquatic organisms (Blanchfield et al., 2021). Specifically, the atmospheric deposition fluxes of Hg over China are referenced from Chen et al. (2019) Atmospheric transport is simulated using the GEOS-Chem chemical transport model (version 9–02; http://geos-chem.org), enabling values for Hg deposition at the 42 study sites (Table S4).

Sensitivity analyses. Model sensitivity analyses were conducted to evaluate the relative importance of key processes on MeHg accumulation in rice and MeHg concentration in paddy water. Specifically, we modeled the reductions of MeHg in both rice and paddy water (%) in response to 10 % decreases in irrigation (mimicking the MeHg input

process), methylation rate (mimicking the Hg methylation process), light interception, and photodegradation rate (k_{PD}).

2.3. Statistical analyses

Differences among treatments were tested using the one-way analysis of variance (ANOVA, SPSS, version 16.0). Posthoc tests were used to identify variances among groups (Tukey, $p < 0.05$).

3. Results and discussion

3.1. Significant MeHG photodegradation in paddy water during rice cultivation

In contrast to the commonly held view that the photodegradation of MeHg in rice-paddy systems is negligible (Alpers et al., 2014), our study, for the first time, reported significant MeHg photodegradation in paddy water during rice growth. The photodegradation rates of MeHg, obtained from a pot experiment simulating rice growing in the field, ranged from 0.007 to 0.0128 h^{-1}. After correcting for photosynthetically active radiation (PAR), we calculated the degradation potentials (i.e., k_{PD}s) to be 8.53 to 12.71 $\times$ 10^{-3} m^2 E^{-1} (Fig. 1a). These values fall within the range reported in surface waters, such as Florida Everglades (10.87 $\pm$ 0.86 $\times$ 10^{-3} m^2 E^{-1}) (Li et al., 2010), Chongqing Jialing River (0.78–12 $\times$ 10^{-3} m^2 E^{-1}) (Sun et al., 2015), and streams (17.95 $\pm$ 0.70 $\times$ 10^{-3} m^2 E^{-1}) (Tai et al., 2017). Photodegradation was found to be the strongest at the tillering stage, followed by the heading and flowering stages (Fig. 1a).

The variation in photodegradation potential during rice growth was mainly attributed to the light interception by rice leaves, as the decreases in k_{PD} matched with the increase in light interception (Fig. 1b). Specifically, rice leaves blocked 0, 3 %, 31 %, and 40 % of solar radiation at different stages in the pot experiment, respectively, wherein 0, 5 %, 30 %, and 44 % reductions in MeHg degradation potential were correspondingly observed (Fig. S1b). Light interception was confirmed in field conditions, where rice leaves blocked 23 %, 67 %, 74 %, and 63 % of light on the 30th, 60th, 90th, and 120th day of rice growth, respectively (Fig. 1b). Light interception was more pronounced in the field compared to the pot experiments, mostly due to differences in rice variety used in the pot experiments (Wufeng 2168 with a height of 95.5 – 102.4 cm) and planted in the field (Zhongkexilu No.4 with a height of ~110 cm) (Data from China Rice Data Center, https://www.ricedata.cn/).

To assess the extent of MeHg photodegradation more comprehensively, degradation rates were determined in laboratory simulation experiments using 42 types of paddy soils collected from 12 provinces of China. Paddy water was sterile-filtrated (0.22-μm) to exclude the impact of microbial demethylation, which is usually considered the primary pathway for MeHg degradation in paddy soils (Wu et al., 2020; Zhou et al., 2020). The observed degradation rates ranged from 0.10 to 0.27 h^{-1} (Fig. 1c), indicating the common occurrence of significant MeHg photodegradation in paddy water. It should be noted that the photodegradation rates were quantified using a closed system, i.e., sealed quartz tubes, which avoided MeHg exchange between pore water and paddy water, as well as interference from rice planting and irrigation. These interferences could possibly change water properties and, subsequently MeHg photodegradation. Further quantification of MeHg photodegradation in real paddy fields and the potential impact of the above-mentioned factors is warranted. Nonetheless, our findings suggest that photodegradation is a crucial yet underestimated pathway of MeHg removal in paddy water.

3.2. Hydroxyl radical-driven MeHG photodegradation in paddy water

By quenching reactive species reported to be involved in MeHg photodegradation (Han et al., 2017), we demonstrated that MeHg photodegradation in paddy water is mainly driven by hydroxyl radicals and hydrated electrons. The quenching of either hydroxyl radicals or hydrated electrons led to significant reductions of MeHg photodegradation by 32–45 % and 30–40 %, respectively, in all examined paddy waters. By contrast, quenching of singlet oxygen or superoxide anions did not result in a significant change in MeHg photodegradation

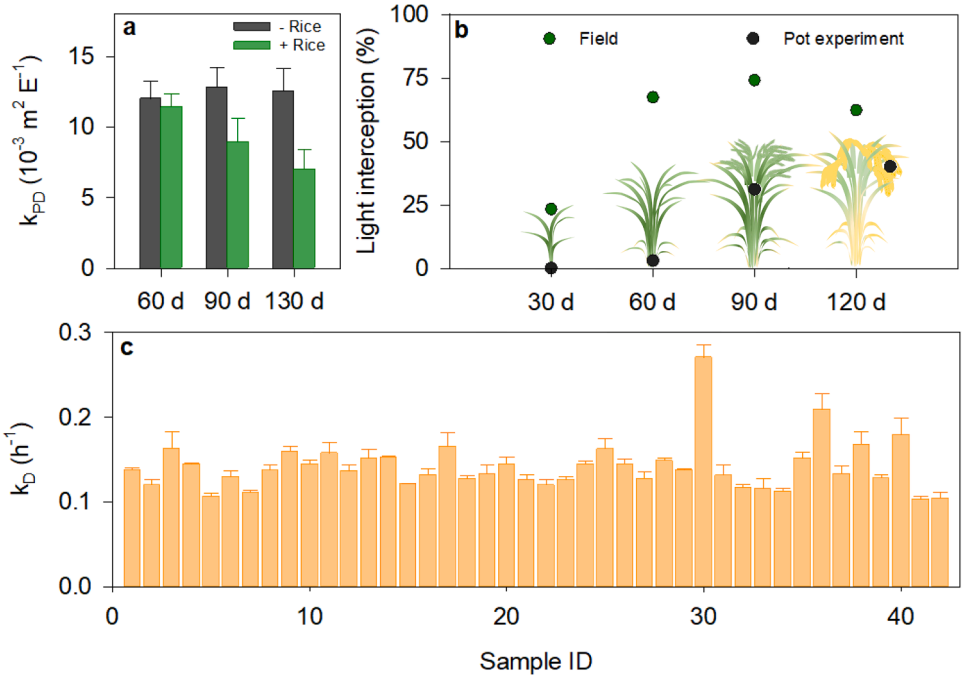


Fig. 1. Methylmercury photodegradation commonly occurs in paddy surface water. (a) The photodegradation rates at different stages of rice growth were obtained from the pot experiments. – Rice and + Rice indicates the treatments without and with rice plants, respectively. (b) The light interception by rice plants throughout the rice growth period in the field. (c) Photodegradation rates in the paddy water of 42 paddy soils in the laboratory simulation experiments. For panels (a) and (c), data are presented as mean $\pm$ standard deviation ($n = 3$).

($p > 0.05$; Fig. 2a).

The dominant role of hydroxyl radicals in MeHg photodegradation was further supported by the higher photodegradation rates observed in waters with lower pH or with UVA radiation (Fig. 2b–c, and Fig. 3a). Specifically, it has been reported that acidic condition favors the generation of hydroxyl radicals (Fischbacher et al., 2017; Lunar et al., 2000), and significant higher generation rate of hydroxyl radicals was observed in waters with lower pH (Fig. S3a). As a result, photodegradation rates were 0.11–0.15 h^{-1} at pH = 4, and declined by 17–24 % at pH = 7 and 35–44 % at pH = 10 in the laboratory simulation experiments (Fig. 2b). In addition, a significant negative relationship was found between the degradation rate and pH of paddy water for 42 soils collected nationwide ($p < 0.05$, Fig. 3a), with the highest degradation rate (0.27 h^{-1}) was found in Yingtan soil with the lowest pH of 4.1 (Fig. 2c). Meanwhile, UVA is involved in the generation of hydroxyl radicals (Kim et al., 2018; Kim and Zoh, 2013), as observed in this study (Fig. S3). When UV (both UVA and UVB) was filtered, the rate of MeHg degradation decreased significantly compared to those quantified under either full spectrum irradiation or with only UVB blocked ($p < 0.05$; Fig. 2c), indicating the dominant role of UVA in facilitating MeHg photodegradation (Lehnherr and St. Louis, 2009).

Despite the key role of hydroxyl radicals in MeHg photodegradation, as evidenced above, the mechanism of hydroxyl radical production in paddy water remains largely unknown. Although DOM and anions are important precursors of hydroxyl radicals, for instance, nitrate account for 24–100 % of the total generation in paddy water (Zeng et al., 2021), no significant positive correlation was observed between photodegradation rate and DOC (Fig. 3b) or anion concentrations (i.e., nitrate, chloride, and sulfate; Fig. S4). Instead, DOM inhibit MeHg photodegradation probably via attenuating light transmission (Klapstein et al., 2018), as evidenced by the significant positive correlation between photodegradation rate and light transmittance in paddy water, as well as the significant negative correlation between transmittance and DOC concentration (Fig. 3c and d).

Therefore, to better elucidate the mechanisms of MeHg photodegradation in paddy water, it is necessary to further explore the contribution of DOM, nitrate, and other possible pathways, such as photo-Fenton reactions (Li and Cai, 2013), to the generation of hydroxyl radicals in paddy water. This is important because paddy water is different from the natural water in lakes and rivers, and is characterized by relatively higher concentrations of DOC (15–74 mg L^{-1}) and nitrate (10–207 mg L^{-1}; Table S1) due to farming activities such as organic/inorganic fertilization. Furthermore, the complex of MeHg with different DOM components has been reported to result in difference in MeHg degradation in natural water (Li et al., 2024; Thompson et al., 2023; Zhang et al., 2022), while how DOM molecular properties and DOM contents impact MeHg degradation in paddy water warrants further study. Meanwhile, the contribution of dissolved iron (Fe), which is abundant in paddy soils and reported to be involved in MeHg photodegradation via the formation of hydroxyl radicals (Hammerschmidt and Fitzgerald, 2010), to MeHg photodegradation also needs future exploration. Additionally, despite the contribution of hydrated electrons to MeHg photodegradation observed in the quenching experiments (Fig. 2a), the mechanism remains unclear. One possible pathway is that hydrated electrons mediate MeHg degradation by forming hydroxyl radicals. Thus, hydrated electrons can be captured by dissolved oxygen to form superoxide anions, which are then converted into hydrogen peroxide and, eventually, hydroxyl radicals (Tai et al., 2012).

3.3. *Photodegradation-induced reductions in MeHG accumulation and MeHG export*

The exchange of MeHg between paddy water and pore water is an important flux in a rice-paddy system (Windham-Myers et al., 2014);

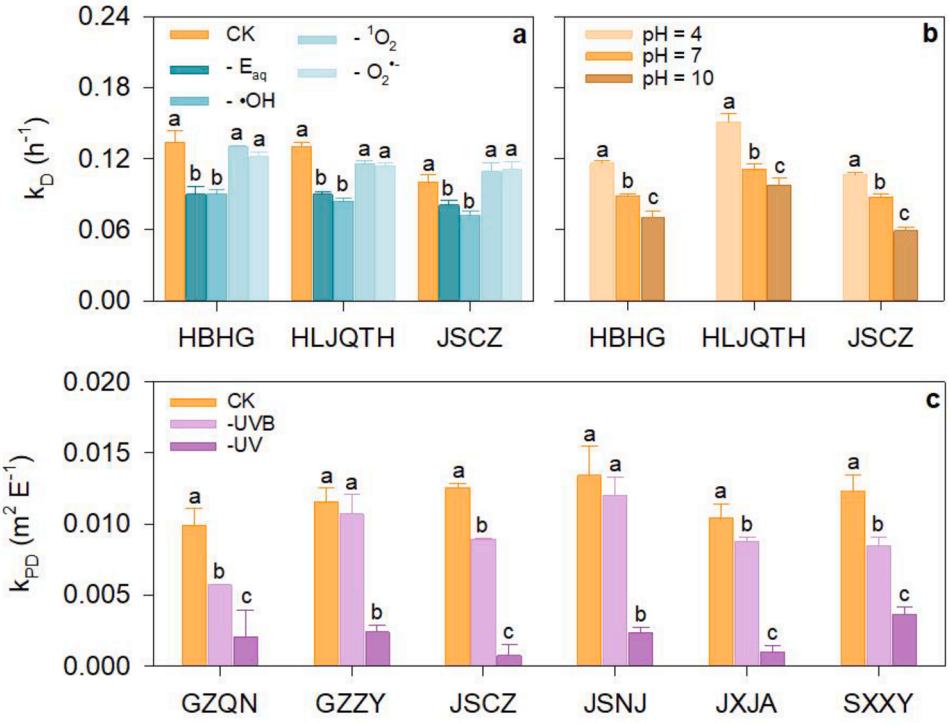


Fig. 2. The effects of active species (a), pH (b), and light quality (c) on MeHg photodegradation in paddy water. For panels (a) and (b), data were obtained from laboratory simulated experiments under 500 W xenon light. For (c), data were obtained from photodegradation experiments under natural sunlight. In active species-quenching experiments, 10 mM isopropyl alcohol, 10 mM dimethyl sulfoxide in nitrous oxide, 10 mg L^{-1} superoxide dismutase, and 10 mM tetrahydrofuran was used to scavenge hydroxyl radicals, hydrated electrons, superoxide anions, and singlet oxygen, respectively. Detailed soil information and experimental design can be found in Tables S1–S2 in Supplementary Information (SI). Different letters (i.e., $a - c$) above the bars indicate significant differences among treatments (one-way ANOVA, $p < 0.05$). Data are shown as the mean ± standard deviation ($n = 3$).

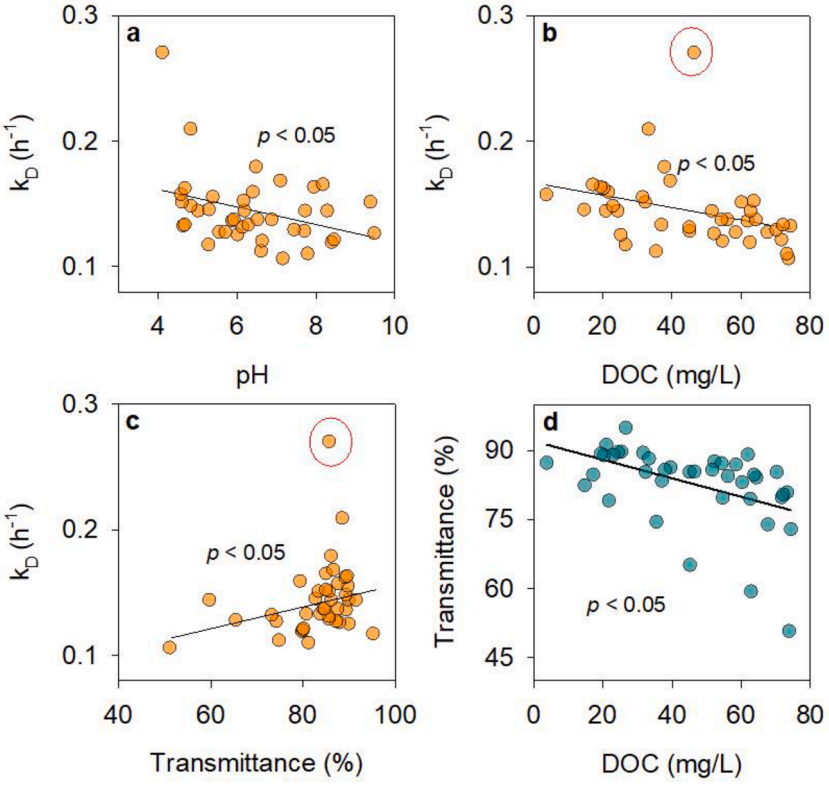


Fig. 3. The correlations between photodegradation rate (k_D) and pH (a), DOC concentration (b), or transmittance of paddy water (c), and between transmittance and DOC (d). For panels (b) and (c), the data from Yingtan soil (circled in red) was excluded from the linear regression due to its strong acidity (pH = 4.1) might suppress all factors in controlling MeHg photodegradation.

therefore, the significant MeHg photodegradation occurring in paddy water would lead to changes in MeHg concentration of pore water and the subsequent MeHg absorption by rice roots (Rothenberg and Feng, 2012). Here, a modified rice paddy biogeochemistry model (RPBM; details in Section 2 of Methods) was applied to quantify the impact of MeHg photodegradation on MeHg removal within rice-paddy systems, considering its good performance in predicting MeHg concentrations in soils (Fig. S5) and rice grains (Fig. 4a and b; Supplementary Text 2).

Model results suggest that photodegradation reduces MeHg accumulation in rice grains by 11.3 % on a national scale in China (Fig. 4c, - k_D versus Current). The variability of predicted reduction among provinces was minimal (ranging from 11.1 to 11.5 %), despite a larger variation in k_{PD}s (0.23 to 0.36 d^{-1}; Table S3). By contrast, rice MeHg would decrease by only 1.2 % if there was no light interception by rice plants (Fig. 4c, + max k_D versus Current). These results suggest that photodegradation in paddy water reduces MeHg accumulation in rice, but further intensifying photodegradation would have only a minor effect in mitigating MeHg risks of rice consumption. The relatively narrow variability in rice MeHg accumulation under different scenarios (Fig. 4c) is mainly attributed to the fact that rice roots absorb MeHg directly from the pore water (Rothenberg and Feng, 2012) instead of paddy water; thus, photodegradation only indirectly affects MeHg accumulation via

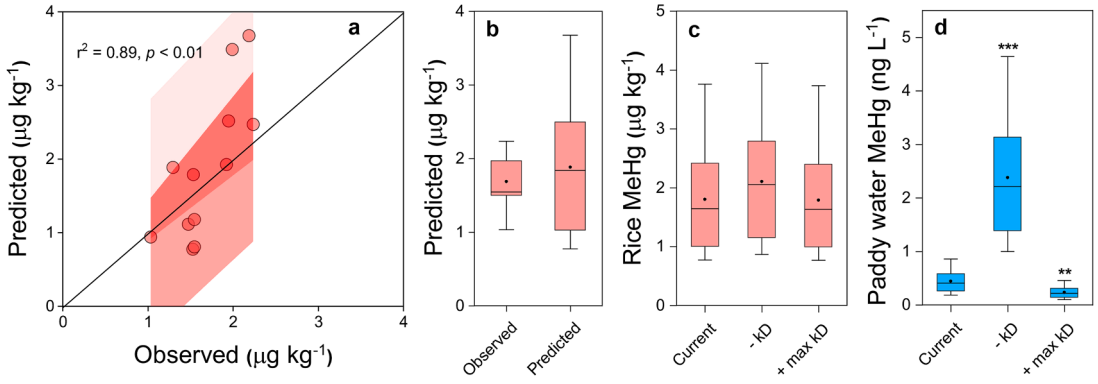


Fig. 4. Predictions of MeHg levels in rice grains ($a - b$) and MeHg concentrations in rice grains (c) and paddy water (d) under different scenarios using the modified rice paddy biogeochemical model. Twelve provinces in China were selected due to their vast rice-planting areas. The concentration of MeHg in paddy water was calculated by dividing the output MeHg (kg) in surface water by the water mass, assuming that the surface water has a depth of 10 cm(Global Rice Science Partnership, 2013) and a density of 1 kg L^{-3}. For panel (a), the pink and light pink shaded areas represent the 95 % confidence interval and 95 % prediction interval, respectively. For panels (b) to (d), the box, whisker, line, and dot indicate a 25–75 % range, 1.5 IQR range, median, and mean values, respectively. ** and *** suggest a significant level of $p < 0.01$ and 0.001, respectively. For panels (c) and (d), - k_D and + max k_D represent scenarios wherein there was no photodegradation or light interception by rice plants, respectively.

MeHg exchanges between these two pools, i.e., infiltration and diffusion (Kwon et al., 2018). Results of sensitivity analyses also support that MeHg accumulation in rice is less responsive to photodegradation, as rice MeHg is ~12 − 28-fold less sensitive to changes in MeHg photodegradation (including light interception and photodegradation rate) than it is to Hg methylation (Fig. S6a and Supplementary Text 3).

Notably, photodegradation significantly reduces MeHg concentration in paddy water. The predicted MeHg in paddy water ranges from 0.18 to 0.86 ng L^{-1}, with a national average of 0.44 ng L^{-1} (Fig. 4d), which is comparable to reported values in surface paddy water (~0.9 ng L^{-1} in Rothenberg et al. (2012) and < 0.4 ng L^{-1} in Tanner et al. (2018)). Photodegradation reduces MeHg content in surface paddy water by 81.5 % (Fig. 4d, - k_D versus Current), which would further decrease by 47 % without light interception by rice plants (+ max k_D versus Current in Fig. 4d). Results of sensitivity analyses also evidence the critical role of photodegradation in controlling MeHg concentrations in paddy water, as MeHg contents in surface paddy water would decrease by 9.48 %, −8.87 %, 7.53 %, and −0.51 % in response to a 10 % reduction in methylation rate, photodegradation rate (k_{PD}), light interception, and irrigation input, respectively (Fig. S6b; Supplementary Text 3). Therefore, the high MeHg content in paddy surface water, either without or with weaker photodegradation, would result in an elevated MeHg accumulation in wildlife inhabiting rice-paddy systems (Windham-Myers et al., 2014), including fish, spiders, frogs, birds, etc.

As hydrologically connected with streams, rivers, and lakes, rice-paddy systems, if without photodegradation, would become a significant MeHg source to the downstream aquatic systems via runoff (Brigham et al., 2009). By multiplying the predicted national-average MeHg content in paddy surface water (2.38×10^{-6} kg ha^{-1}), the national total rice planting area (30 million ha in 2020; Chinese Statistical Yearbook, 2021), and the annual rate of runoff (2.5 − 12 yr^{-1}) (Bachand et al., 2014; Kim et al., 2014), we estimate that without MeHg photodegradation, rice-paddy systems would deliver an annual mass of 178−856 kg MeHg to downstream habitats in China. It is important to note that the amount of MeHg output does not necessarily equate to the input to the receptors because MeHg could be lost during transportation, for instance, via photolysis, microbial demethylation, adsorption, and absorption by biota. Unfortunately, we are also unable to assess the amount of MeHg loss due to the lack of related data. Similarly, the spatial and temporal differences in the intensity of solar radiation have not been considered in the rice paddy model. Furthermore, the ignorance of artificial drainage during rice growth in the rice paddy model may lead to overestimations. By contrast, rice is assumed to be planted once a year in the model, resulting in underestimations of MeHg annual export, as rice is planted twice in major planting areas such as Hunan and Jiangxi Province. Nonetheless, this proportion of MeHg is significant, as it accounts for up to 16 % (7.3 − 33.6 % on a provincial scale in China; Table S4) of atmospheric THg deposition, which is recognized as a major source of Hg in aquatic systems (Blanchfield et al., 2021).

4. Conclusion

By revealing that MeHg photodegradation is a significant MeHg removal pathway in paddy water throughout rice growth, this study demonstrates that this overlooked natural process provides a crucial ecosystem service by reducing MeHg levels in both rice grains and the runoff to downstream habitats. Results obtained from this study challenge the traditional view that MeHg photodegradation is negligible in rice-paddy systems because of light attenuation by plant leaves, advancing our understanding of Hg biogeochemistry in wetland systems. Unfortunately, we are unable to estimate the extent to which photodegradation in rice-paddy systems prevents MeHg exposure to humans via the consumption of aquatic products due to the lack of data on the proportions of paddy runoff to downstream aquatic systems. Nonetheless, this study quantifies several key aspects of the otherwise hidden role that MeHg photodegradation in rice-paddy systems plays in reducing MeHg bioaccumulation in downstream waters, as well as provides novel insights into Hg mitigation in paddy-connected aquatic systems, e.g., by intensifying this natural process in rice-paddy systems via cultivating dwarf rice (Jeon et al., 2000).

CRediT authorship contribution statement

Huan Zhong: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Conceptualization. **Hangyu Zhou:** Investigation, Formal analysis, Data curation. **Yanbin Li:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Formal analysis, Conceptualization. **Chengjun Li:** Writing – review & editing, Visualization, Funding acquisition. **Martin Tsz-Ki Tsui:** Writing – review & editing. **Carl P.J. Mitchell:** Writing – review & editing. **Yang Zhou:** Writing – review & editing. **Yanan Yang:** Investigation, Data curation. **Long Chen:** Writing – review & editing, Software, Methodology. **Hongqiang Ren:** Resources. **Wenli Tang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.watres.2024.121332.

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